Conformal covariance of the supercritical Liouville quantum gravity metric

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Abstract

Let $\xi > \xi_{\mathrm{crit}}$, and let $Q = Q(\xi)$ be the LFPP distance exponent. We prove that the supercritical Liouville quantum gravity metric respects the coordinate change $h \mapsto h \circ \phi^{-1} + Q \log |(\phi^{-1})'|$ under a fixed conformal map ϕ. The result holds for a zero-boundary Gaussian free field plus a random continuous function on any connected Greenian domain, and the equality is simultaneous in all endpoint pairs, including singular points. The main difficulty is that the metric is not continuous in the Euclidean topology. We overcome it by joining recovery paths to their prescribed endpoints through short surrounding loops. A local comparison along geodesics then improves a deterministic bi-Lipschitz bound to equality.

1 Introduction

1.1 Overview

A Liouville quantum gravity surface is described by a domain together with a random generalized function. Different pairs can describe the same surface: if $\phi : U \to V$ is conformal, the coordinate change is

$$h^{\phi} = h \circ \phi^{-1} + Q \log |(\phi^{-1})'|. \tag{1}$$

A metric associated with this surface should therefore give the same distance before and after changing coordinates. The purpose of this paper is to establish this property for the supercritical LQG metric.

Throughout we fix $\xi > \xi_{\mathrm{crit}}$ and write $Q = Q(\xi) \in (0, 2)$. We use one normalization of the strong whole-plane metric constructed and characterized in [1]. Unlike the subcritical metric, this metric has a dense set of singular points, each at infinite distance from every other point [3]. Distances are lower semicontinuous as functions of their Euclidean endpoints, but they are not Euclidean continuous. Consequently, approximating a conformal map by its derivative does not immediately give a comparison at fixed endpoints.

Our argument follows the comparison strategy of [2]. At small scales a conformal map becomes linear, and the transformed field approaches the corresponding affine transform of the original field. In the supercritical setting the resulting metric convergence allows the endpoints of a recovering path to move. We show that this is enough when the prescribed endpoints lie on suitable geodesic segments: short loops near those endpoints provide connectors whose length tends to zero. A covering by annuli containing such segments then yields exact equality of the two metrics.

1.2 The metric on a domain and the main result

We first specify the metric to which the coordinate-change formula applies. A domain is a nonempty connected open subset of $\mathbb{C}$. It is called Greenian when it admits a Green function, so that its zero-boundary GFF is defined. No regularity of the boundary is assumed. If D is an extended metric, $D(x, y; W)$ denotes its internal distance in W, obtained by infimizing the D-lengths of paths in W.

Definition 1.1 (The whole-plane metric)**.** We fix an assignment $h \mapsto D_h$ satisfying the strong LQG metric axioms with parameters ξ, Q. For a whole-plane GFF plus a continuous function, D_h is almost surely a lower-semicontinuous extended length metric. Its internal metric on an open set W is determined by $h|_W$. For every continuous f,

$$D_{h+f} = e^{\xi f} \cdot D_h, \qquad (e^{\xi f} \cdot D)(x, y) = \inf_{P:x\to y} \int_0^{\mathrm{len}(P;D)} e^{\xi f(P(t))}\, \mathrm{d}t, \tag{2}$$

where paths are parametrized by D-length. It also satisfies translation and dilation covariance. In particular, for each fixed complex affine map $\psi(z) = az + b$, $a \neq 0$,

$$D_{h\circ\psi+Q\log|a|}(z, w) = D_h(az + b, aw + b). \tag{3}$$

The complex affine form includes rotation invariance, which follows from uniqueness; see [1, Proposition 1.9].

The existence and uniqueness theorem of [1] says that this assignment is determined up to a deterministic multiplicative constant. We choose this constant once, for all fields and domains. All equalities of metrics below mean equality at every pair of points on a single probability-one event. Continuous Weyl scaling is used on its simultaneous event, and therefore also applies to random continuous functions which depend on the field.

Definition 1.2 (The metric on a Greenian domain)**.** Let $h = h_0^U + f$ on a connected Greenian domain U, where h_0^U is a zero-boundary GFF and f is a measurable random continuous function. Realize h, if necessary on an enlarged probability space, as

$$h = H|_U + F, \tag{4}$$

where H is a whole-plane GFF plus a continuous function and $F \in C(U, \mathbb{R})$. Set

$$D_h^U(x, y) = \inf_{P:x\to y,\ P\subset U} \int_0^{\mathrm{len}(P;D_H)} e^{\xi F(P(t))}\, \mathrm{d}t. \tag{5}$$

The infimum of an empty family is $+\infty$, and constant paths have length zero.

Section 2.1 proves that the realization exists, that (5) does not depend on its choice, and that the metric is a measurable function of h. Thus it is defined on the original probability space of h. The function f may depend on h_0^U and may be unbounded near ∂U.

Theorem 1.3 (Conformal covariance)**.** *Let U be a connected Greenian domain, let $\phi : U \to V$ be a deterministic conformal bijection, and let h be a zero-boundary GFF on U plus a random continuous function. Almost surely,*

$$D_h^U(z, w) = D^V_{h\circ\phi^{-1}+Q\log|(\phi^{-1})'|}(\phi(z), \phi(w)), \qquad z, w \in U. \tag{6}$$

Both metrics are measurable functions of their respective fields and are defined on the original probability space of h.

In particular, the metric respects the equivalence relation (1). Formula (6) includes singular endpoints and infinite distances. Its multiplicative constant is exactly one, since both sides use the same whole-plane normalization. The conformal map is fixed before the probability-one event is chosen.

For precision, the measurable functions in the theorem are chosen for the law of the field under consideration. The construction has the following useful countable form.

Corollary 1.4. *Fix U, V. For any prescribed countable family of fields h_i as in Theorem 1.3 and conformal maps $\phi_i : U \to V$, one can choose the same two measurable functions*

$$\Phi_U : \mathcal{D}'(U) \longrightarrow [0,\infty]^{U\times U}, \qquad \Phi_V : \mathcal{D}'(V) \longrightarrow [0,\infty]^{V\times V}$$

to define all the source and target metrics. For each i, formula (6) holds almost surely on the probability space of h_i. On every extension carrying a realization of either field, these functions agree almost surely with (5).

The countable family is specified before the functions are chosen. Neither statement requires one version valid for all field laws, or a common event for all conformal maps.

1.3 Outline and notation

For most of the proof, h is a whole-plane GFF restricted to U. We compare

$$d(z,w) = D_h(z,w;U), \qquad e(z,w) = D^V_{h^\phi}(\phi(z),\phi(w)). \tag{7}$$

Section 2.3 proves a deterministic bi-Lipschitz bound between d and e. Section 3 identifies their common limit after zooming in around a point. In Section 4 we use short loops to recover prescribed endpoints and obtain a comparison with factor $1+\eta$ along regular geodesic segments. Sections 5 and 6 turn this local comparison into $e \le d$. Applying the same argument to ϕ^{-1} gives equality.

We use the established whole-plane LQG metric theory, including annular estimates and the GFF Markov property. We state the precise estimates when they enter. The conformal comparison, endpoint recovery, and improvement to constant one are proved below; no conformal-covariance theorem is assumed.

We write $B_r(z)$ for a Euclidean ball and $A_{r,R}(z) = B_R(z) \setminus \overline{B}_r(z)$. Distances *across* an annulus are infima over crossing paths, and distances *around* it are infima over loops separating its boundary components. Boundary endpoints are allowed. Whenever an internal metric has boundary endpoints, its carrier is the closed annulus. The notation $K \Subset U$ means that K is compact and contained in U. Finally,

$$m_{z,r} = r^{\xi Q} e^{\xi h_r(z)},$$

where $h_r(z)$ is the circle average of the whole-plane field.

2 Domain metrics and a first comparison

2.1 Constructing the metric on a domain

We record the details behind Definition 1.2. They will also let us work with local fields without imposing boundary conditions on a continuous correction.

Lemma 2.1. *The construction in Definition 1.2 is well defined. For each prescribed law of h, it has a measurable version depending only on h. It satisfies locality, the length-space property, restriction to internal metrics, and continuous Weyl scaling. These assertions also hold with one choice of measurable versions for any prescribed countable family of laws.*

Proof. First consider a disk $B \Subset B' \Subset U$. The Markov decomposition writes the field on B' as a zero-boundary GFF plus a continuous function. Couple this zero-boundary part to that of a whole-plane GFF H. The difference between the two fields on B' is continuous. Multiplying it by a cutoff which is one near $\overline{B}$ gives a continuous function on $\mathbb{C}$, and hence defines a metric on B by whole-plane Weyl scaling and locality.

We explain why these local choices agree. Suppose two whole-plane fields plus continuous functions agree on an open neighborhood W. Choose between the two fields by an independent fair coin. This mixture is again a whole-plane GFF plus a random continuous function, after absorbing normalization constants into that function. Locality for the mixture gives one measurable function of the restriction to W for its internal metric. Since each component has positive probability, this same function represents the internal metric in both components. Thus the internal metrics agree on the original coupling whenever the fields agree on W. Applying this observation to the cutoff fields proves consistency on overlaps.

Choose a countable relatively compact cover by disks. The consistent local metrics assign lengths to paths, and subdivision makes this assignment independent of the cover. Infimizing these lengths gives (5). Every path has compact image in U, so only finitely many disks are needed for any given path. This also proves the internal restriction identity and Weyl scaling. An unbounded continuous correction causes no problem, because it is bounded on each such compact image.

A realization on the full domain exists as well. Take a reference whole-plane Markov decomposition $H|_U = Z + \mathfrak{h}$, where Z has the zero-boundary GFF law and $\mathfrak{h}$ is harmonic on U. Sample H conditionally on $Z = h_0^U$, on a probability-space extension. Then $h = H|_U + (f - \mathfrak{h})$ there. This conditional sampling is over the standard Borel space of distributions; it does not require the original probability space to be standard Borel. Repeating the overlap argument shows that any two realizations give the same metric on a common extension.

For measurability, exhaust U by compact sets K_n with $K_n \subset \operatorname{int} K_{n+1}$. The internal metric on each K_n is lower semicontinuous, by Lemma 2.2 below, and is determined by the field on a neighborhood of K_n. Its epigraph is a random closed set, so it has a countable measurable description. Locality and measurable factorization give a function of h for each of these compact internal metrics. Their increasing union determines all path lengths and hence the domain distance: every path lies in some K_n, and the distance is the infimum of the corresponding compact internal distances. This proves the assertion simultaneously in all endpoints.

For countably many laws, apply this construction to a mixture $\sum_i w_i \mu_i$, where all $w_i > 0$ and $\sum_i w_i = 1$. The almost-sure identities under the mixture hold under every component. The same construction therefore supplies one function for all the specified laws. Further details concerning extensions and descent to the original space are given in Appendix A.1. □

The transformed field h^ϕ belongs to the same class: conformal invariance sends h_0^U to a zero-boundary GFF on V, and $f \circ \phi^{-1} + Q \log |(\phi^{-1})'|$ is continuous on V. In particular, the target metric in Theorem 1.3 is defined without assuming the conclusion of that theorem.

2.2 Compactness and annular estimates

We use the following consequences of the whole-plane metric theory: the local Hölder and geodesic properties in [3, Propositions 1.10 and 1.12], the escape estimate in [3, Lemma 3.12], and the annular estimates in [1, Section 2.5]. Finite-length metric paths are Euclidean continuous. On each compact set, the identity from the metric to the Euclidean topology has a Hölder modulus. Nonsingular points can be joined by whole-plane geodesics. The distance between disjoint Euclidean compact

sets is positive. For any fixed nondegenerate round annulus A, the random variables

$$m_{z,r}^{-1} D_h(\text{around}\,(z + rA)), \qquad m_{z,r} D_h(\text{across}\,(z + rA))^{-1} \tag{8}$$

are tight, uniformly in z, r; both annular distances are almost surely positive and finite. These statements concern finite-length paths and annular distances, and do not assert a finite diameter for a Euclidean ball.

We also use the annular iteration estimate [1, Lemma 2.1]. In the form needed here, events measurable from the GFF modulo constants on concentric annuli with a fixed geometric separation have an exponentially small probability of too few successes. If their probabilities are sufficiently close to one, the exponential rate can be made arbitrarily large. If their probabilities have a fixed positive lower bound, there is some positive exponential rate. Enlarging each annulus by a fixed collar only changes the separation factor. This estimate applies to measurable versions of events in the completed field sigma algebra.

Lemma 2.2 (Compact path limits)**.** *Let K be a Euclidean compact subset of a domain carrying one of the metrics above. Paths in K with bounded metric length, parametrized at constant metric speed on $[0, 1]$, have a Euclidean-uniformly convergent subsequence. If $P_n \to P$ uniformly, then*

$$\operatorname{len}(P; D) \le \liminf_n \operatorname{len}(P_n; D).$$

The distance internal to K is Euclidean lower semicontinuous.

Proof. Cover K by finitely many disks where the field differs continuously from a whole-plane GFF. Cutoff extensions and Weyl scaling transfer the local metric-to-Euclidean modulus to each disk. A smaller concentric cover and the positive cost of leaving its larger disks give a common Euclidean modulus for the paths. Arzelà–Ascoli therefore applies.

For a fixed partition $0 = t_0 < \cdots < t_k = 1$, lower semicontinuity of the local distances bounds the sum of the limiting increments by the liminf of the lengths. Subdivide so that each piece lies in a coordinate disk, then take the supremum over partitions. The limit has finite metric length when the liminf is finite; the same increment bound also makes it a metric-continuous path after length parametrization. Finally, apply the argument to near-minimizing paths with convergent endpoints in K. □

2.3 A finite comparison constant

Keep the notation d, e from (7), with h a whole-plane GFF. We begin with a bound which need not be close to one.

Proposition 2.3. *There is a deterministic $C \ge 1$, depending only on the metric parameter, such that almost surely*

$$C^{-1} d(z, w) \le e(z, w) \le C d(z, w), \qquad z, w \in U. \tag{9}$$

The same constant compares path lengths and distances with paths restricted to any common carrier.

Proof. Fix $K \Subset U$ and work on a compact neighborhood of K. Uniformly for z in that neighborhood, the maps

$$F_{z,r}(w) = \frac{\phi(z + rw) - \phi(z)}{r\phi'(z)}$$

converge to the identity in C^∞ on fixed disks as $r \to 0$. The field comparison proved in Lemma 3.1 below, together with affine covariance and Weyl scaling, transfers (8) to the pullback metric e. To

see this without using any metric covariance under ϕ, sandwich the image of each source annulus between round target annuli with slightly smaller and larger radii. A loop in the smaller target annulus pulls back to a source separating loop, and a source crossing must cross an appropriate target subannulus. The deterministic distortions tend to zero. A smooth field normalization differs from the circle normalization by a tight Gaussian constant. Thus both normalized target quantities are tight as well.

It follows that, for every $p < 1$, a deterministic $C_0 < \infty$ can be chosen so that for all small r,

$$\mathbb{P}\left[e(\text{around } A_{r,2r}(z)) < C_0 D_h(\text{across } A_{r/2,r}(z))\right] \geq p. \tag{10}$$

There is an analogous estimate with the metrics interchanged. The constants are uniform on normalized univalent maps on disks whose radii tend to infinity, hence C_0 does not depend on K, U, ϕ. The scale below which the estimate holds may depend on them. The events are local to a fixed annular collar and unchanged by adding a constant to h.

Apply annular iteration at radii between ε^2 and ε, at lattice points of mesh $\varepsilon^2/100$. Choosing p sufficiently close to one makes the failure probability at a lattice point $O(\varepsilon^{10})$. There are $O_K(\varepsilon^{-4})$ relevant points. With probability $1 - O_K(\varepsilon^6)$, every point of K lies in $B_{r/2}(z)$ for a ball satisfying (10).

We give the deterministic chaining argument, since successive surrounding loops need not intersect. Let P be a finite d-length path. Use a target metric confined to a fixed compact neighborhood of P, so that it is lower semicontinuous. At time t, choose a good ball containing $P(t)$ in its inner half and a target loop ℓ surrounding its inner ball. Its target diameter is at most $C_0 c$, where c is the source crossing cost from radius $r/2$ to r.

For a compact connected set L, let $\mathrm{Fill}(L)$ be L together with its bounded complementary components. Maintain a reached set R with the following properties: every point of R is reachable from some point within 4ε of $P(0)$ at cost at most $C_0 \operatorname{len}(P|_{[0,t]}; d)$; and every separating loop whose fill contains $P(t)$ either meets R or also surrounds $P(0)$. Initially take $R = \varnothing$.

If P subsequently meets ℓ, let t' be its first such time. It has paid at least c between t and t'. If ℓ meets R, its entire trace is now reachable by paying its diameter. If it surrounds $P(0)$, any point of ℓ is within 4ε of $P(0)$ and can serve as a new starting point. Adjoin ℓ to R. The fill property persists: a new separating loop containing $P(t')$ either meets ℓ, or contains all of ℓ in its fill; in the latter case apply the previous fill property. The source costs are charged to consecutive disjoint time intervals.

If P never meets the chosen loop again, its terminal point lies in that loop's fill. An already reached intersection point with the loop is within 4ε of the terminal point and can be used as the final endpoint. If the loop also surrounds $P(0)$, the same point can be used at both ends. No unpaid traversal of a terminal loop is needed. The procedure terminates: at a fixed covering scale only finitely many balls are available, and their positive crossing costs have a positive minimum. We have obtained $x_\varepsilon \to P(0)$ and $y_\varepsilon \to P(1)$ with

$$e_K(x_\varepsilon, y_\varepsilon) \leq C_0 \operatorname{len}(P; d),$$

where e_K is the confined target metric.

Choose a sequence of cover scales with summable failure probabilities and use Lemma 2.2 to pass to the exact endpoints. A countable compact exhaustion makes the argument simultaneous for every finite-length path. Infimizing gives $e \leq C_0 d$. The reverse loop-to-crossing estimate gives $d \leq C_0 e$ by the same argument. Comparing the sums over every partition of a path proves the length comparison, and infimizing those lengths proves the last assertion. □

3 Comparison near a linear map

We next identify the metric seen after a small conformal change of coordinates. This part of the proof concerns convergence of distance functions, and makes no assertion about convergence at prescribed endpoints.

3.1 A field comparison

Fix a smooth radial function $\rho \geq 0$ supported in the unit disk with $\int \rho = 1$. Normalize the whole-plane GFF by $(h, \rho) = 0$. For $a \in \mathbb{C} \setminus \{0\}$ put

$$\rho_a(w) = |a|^{-2}\rho(w/a), \qquad h^a(w) = h(a^{-1}w).$$

Thus $(h^a, \rho_a) = 0$. For probabilistic convergence arguments we realize the fields in H^{-s}_{loc}, with any fixed $s > 0$. GFFs and their continuous perturbations belong locally to this space almost surely. Its usual countable exhaustion gives a Polish space.

Lemma 3.1. *Suppose U_n eventually contains every compact subset of $\mathbb{C}$, and $\psi_n : U_n \to V_n$ are conformal bijections with $\psi_n(0) = 0$ and $\psi_n'(0) \to a \neq 0$. On every compact set,*

$$\psi_n \longrightarrow a\,\mathrm{id}, \qquad \psi_n^{-1} \longrightarrow a^{-1}\mathrm{id}$$

in C^∞. Moreover,

$$h \circ \psi_n^{-1} \longrightarrow h^a \qquad \text{almost surely as distributions,} \tag{11}$$

$$h \circ \psi_n^{-1} - (h \circ \psi_n^{-1}, \rho_a) \longrightarrow h^a \qquad \text{locally in total variation of laws.} \tag{12}$$

Proof. Apply the distortion theorem on disks $B_R \subset U_n$, and then let $R \to \infty$. It gives the first convergence locally uniformly; Cauchy's formula gives all derivatives. The inverse maps have the same property, and their domains also exhaust $\mathbb{C}$. For a smooth compactly supported test function g,

$$(h \circ \psi_n^{-1}, g) = (h, |\psi_n'|^2(g \circ \psi_n)).$$

The test functions on the right converge with all derivatives and have a common compact support. This proves (11). Approximating h locally in H^{-s} by smooth functions, and using the uniform local bounds for pullback by these diffeomorphisms, also gives convergence in H^{-s}_{loc}.

Here are the local Gaussian details for (12); they also explain the use of the smooth normalization. Choose a disk B_R containing the compact set in question and the support of ρ_a, and a much larger disk B_L. A GFF restricted to B_L, viewed modulo constants, has a decomposition into an independent zero-boundary field and a harmonic field. Recenter both summands against ρ_a. The zero-boundary summand is the same for any of the fields being compared, while the recentered harmonic summand has small Dirichlet energy on B_{2R} as $L \to \infty$.

For completeness, this last estimate follows directly from the disk Green function. For a whole-plane field, the covariance of the harmonic part after subtracting its value at the center is

$$-\log\left|1 - \frac{z\overline{w}}{L^2}\right|, \qquad |z|, |w| < L.$$

On B_{2R} this kernel and its mixed derivatives are $O_R(L^{-2})$. Integrating against ρ_a gives the same bound after recentering against ρ_a. The mean-square C^1 bound on a smaller disk follows from the harmonic mean-value formula. For $h \circ \psi_n^{-1}$ the covariance, modulo terms depending on only one

variable, is $-\log|\psi_n^{-1}(z)-\psi_n^{-1}(w)|$. On B_L, convergence of the maps and their derivatives gives the same harmonic estimate with an additional $o_n(1)$ term, for each fixed L. One may first carry out the decomposition on a larger disk and restrict to B_L, so every boundary used here lies strictly inside V_n.

Multiply the recentered harmonic part by a smooth cutoff equal to one on B_R and supported in B_{2R}. Its expected Dirichlet energy is $O_R(L^{-2})+o_n(1)$. A deterministic Cameron–Martin shift g changes the zero-boundary GFF law by total variation at most a constant times $\|g\|_\nabla$, by the Gaussian relative-entropy formula and Pinsker's inequality. Conditional on the harmonic part, this applies to the independent zero-boundary field. Integrating the conditional bound and using Cauchy–Schwarz gives an error $O_R(L^{-1})+o_n(1)$. Recentring is a measurable map and cannot increase total variation. Both normalized fields are therefore within this error of the same normalized zero-boundary field on B_L, restricted to B_R. First let $n\to\infty$ and then $L\to\infty$. This proves (12). □

This is the field comparison used in [2, Lemma 2.4], written with the normalization transported by $w\mapsto aw$. The test function in the target has scale $|a|$.

We will apply this comparison to measurable metric functions. The following elementary fact is useful because such functions need not be continuous.

Lemma 3.2. *Let X_n, X take values in a Polish space and be defined on one probability space. If $X_n\to X$ in probability and their laws converge in total variation, then $F(X_n)\to F(X)$ in probability for every measurable map F into a separable metric space.*

Proof. By Lusin's theorem, for each $\delta>0$ there is a compact set K with $\mathbb{P}[X\in K]>1-\delta$ on which F is continuous. Total variation gives $\liminf_n \mathbb{P}[X_n\in K]\ge 1-\delta$. On K the function F is uniformly continuous. Convergence in probability of X_n to X now bounds the failure probability for the conclusion by $2\delta+o(1)$. Let $\delta\to 0$. □

3.2 Confining paths

For nonnegative lower-semicontinuous functions we use local epigraph convergence. Thus $f_n\to f$ means that

(i) $f(x)\le\liminf_n f_n(x_n)$ whenever $x_n\to x$;

(ii) for every x there are $x_n\to x$ with $f_n(x_n)\to f(x)$.

Here x is a pair of Euclidean endpoints. Closed epigraphs give a metrizable separable state space; one can use the usual hyperspace of closed sets on a countable compact exhaustion. We use the ambient hyperspace, rather than asserting that length metrics form a closed subspace of it.

There are two simple continuity observations. If D is a lower-semicontinuous metric on a compact carrier and $e^{-a_n}D\le D_n\le e^{a_n}D$ with $a_n\to 0$, then $D_n\to D$ in the epigraph topology. The lower bound follows from lower semicontinuity and the upper bound uses fixed endpoints. Also, if $f_n\to f$ and ψ_n,ψ_n^{-1} converge locally uniformly to a homeomorphism and its inverse, then pullback transports epigraph convergence. For the recovery condition, pull the recovering endpoints back by ψ_n^{-1}.

Lemma 3.3 (A confinement event)**.** *Fix $R<\infty$ and $\delta>0$. There is $S>4R$ such that, with probability at least $1-\delta$, all whole-plane distances between points of B_R equal the corresponding distances with paths confined to $\overline{B}_S$. The event can be chosen local to B_{2S} and stable under sufficiently small uniform Weyl perturbations there.*

Proof. The distance around $A_{2R,3R}$ is finite almost surely. The distance across $A_{S/2,S}$ tends to infinity in probability: by affine covariance it is $S^{\xi Q}e^{\xi h_S(0)}$ times a positive random variable with a fixed law. Since $h_S(0) = o(\log S)$ in probability and $Q > 0$, the assertion follows from tightness of the reciprocal of that variable.

Choose S so that, with probability at least $1 - \delta$, there is a separating loop $L \subset A_{2R,3R}$ with

$$\operatorname{len}(L; D_h) < \tfrac{1}{4} D_h(\text{across } A_{S/2,S}). \tag{13}$$

A path between points of B_R which exits B_S meets L before and after that excursion. Replace the intervening piece by an arc of L. The cost strictly decreases, and the replacement lies in B_S. Taking infima proves the asserted equality, including the case when no finite path exists.

The loop and crossing tests are local. Under a Weyl perturbation g with $\|g\|_\infty \le a$, the ratio of their costs changes by at most $e^{2\xi a}$. The strict margin in (13) therefore persists for all sufficiently small a. □

Proposition 3.4 (The tangent metric)**.** *Under the hypotheses of Lemma 3.1, with the domains U_n Greenian, choose $R_n \to \infty$ with $\overline{B}_{2R_n} \subset U_n$. Let*

$$d_n = D_h(\cdot, \cdot; \overline{B}_{R_n}), \qquad e_n = \psi_n^* D_{h^{\psi_n}}^{V_n}(\cdot, \cdot; \psi_n(\overline{B}_{R_n})),$$

extended by $+\infty$ outside their carriers. Jointly with h,

$$(d_n, e_n) \longrightarrow (D_h, D_h) \tag{14}$$

in probability for local epigraph convergence.

Proof. First fix a compact target disk and a slightly larger neighborhood. Use Lemma 2.1 to choose one local metric function for all the normalized field laws in Lemma 3.1, including the limit law. Its values on a closed disk are lower semicontinuous compact internal metrics. Lemma 3.2 therefore gives convergence in probability of these metrics, jointly with the original field.

The constant removed in (12) tends to zero almost surely, by (11). The remaining continuous correction satisfies

$$Q \log |(\psi_n^{-1})'| + Q \log |a| \longrightarrow 0$$

uniformly on compact sets. Weyl scaling and the multiplicative comparison above show that the compact internal target metrics converge to those for $h^a - Q \log |a|$.

Now fix an inner disk B_R. Lemma 3.3 supplies, with arbitrarily high probability, a larger deterministic disk which computes all its distances. Its margin event transfers to the normalized transformed fields by total variation, and survives the vanishing Weyl correction. Eventually $\psi_n(B_{R_n})$ contains this larger target disk. Distances confined to $\psi_n(\overline{B}_{R_n})$ therefore agree with the fixed confined distances on the inner target disk, except on an event of arbitrarily small probability. The same is true for the source metrics.

Pull the target convergence back by ψ_n. The affine identity (3) identifies its limit with D_h. Taking a countable exhaustion and then letting the confinement error tend to zero proves (14). Both coordinates converge in probability on the original coupling, which also proves the joint assertion. □

The same conclusion in law applies when h is replaced at step n by a field h_n with the same normalized GFF law. Indeed, the joint law at step n is obtained from that one field law by the same field and metric functions. A positive-weight mixture supplies common versions for this countable list of functions. No coupling of the successive h_n is needed.

4 Recovering prescribed endpoints

Epigraph convergence gives good approximating endpoints, but not necessarily the endpoints one wants to compare. The next lemma is the step that replaces Euclidean continuity of the metric.

Lemma 4.1 (Endpoint recovery)**.** *Let d_n, e_n be extended length metrics on sets eventually containing every compact subset of $\mathbb{C}$, with Euclidean-continuous finite-length paths. Suppose $C^{-1}d_n \le e_n \le Cd_n$ and both sequences converge locally in the epigraph topology to the same metric D. Let $u_n \to u$, $v_n \to v$, where $u \neq v$, and suppose $L_n = d_n(u_n, v_n) \to L < \infty$.*

Assume that for some $M, \theta > 0$ and $s_j \downarrow 0$, for $x_n = u_n, v_n$, every fixed j, and all sufficiently large n,

$$\begin{aligned} d_n(x_n, \partial B_{s_j}(x_n)) &\le M s_j^\theta, \\ d_n(\text{around } A_{s_j,\sqrt{s_j}}(x_n)) &\le M s_j^\theta. \end{aligned} \tag{15}$$

Then $\limsup_n e_n(u_n, v_n) \le D(u, v) \le L$.

Proof. The lower-bound condition gives $D(u, v) \le L$. Choose recovery endpoints $a_n \to u$, $b_n \to v$ for e_n, and paths Q_n between them with lengths tending to $D(u, v)$.

Fix a small outer scale s_k. Next choose a smaller scale s_j with $\sqrt{s_j} < s_k/2$. For large n, a_n lies inside $B_{s_j}(u_n)$ and b_n lies outside $B_{s_k}(u_n)$. Choose a separating loop ℓ_n in $A_{s_j,\sqrt{s_j}}(u_n)$ with d_n-length at most $Ms_j^\theta + o(1)$, and a path from u_n to $\partial B_{s_k}(u_n)$ of d_n-length at most $Ms_k^\theta + o(1)$, stopped at its first exit. Both this path and Q_n meet ℓ_n. By $e_n \le Cd_n$, one can connect u_n to the intersection of Q_n with ℓ_n at e_n-cost at most

$$CM(s_k^\theta + s_j^\theta) + o(1).$$

Do the same at v_n, with disjoint endpoint neighborhoods. Use the first source-side intersection of Q_n and the last target-side intersection. Their order is correct since the two loops separate disjoint endpoint neighborhoods. The middle portion of Q_n and the two connectors give

$$\limsup_n e_n(u_n, v_n) \le D(u, v) + 2CM(s_k^\theta + s_j^\theta).$$

Let $j \to \infty$ with k fixed, then let $k \to \infty$. The use of two scales ensures that the radial connector reaches the entire loop; its cost is included in the estimate. □

Let $\Delta = \{2^{-j} : j \in \mathbb{N}\}$. Fix $\alpha \in (3/4, 1)$ and $\theta > 0$. Call $u \in \partial B_{\alpha r}(z)$, $v \in \partial B_r(z)$ a regular pair if there is a whole-plane D_h-geodesic between them contained in the closed annulus $\overline{A}_{\alpha r,r}(z)$, if

$$D_h(u, v) \le (1 - \alpha)^{-1} m_{z,r}, \tag{16}$$

and if, for $x = u, v$ and every $\delta \in \Delta$ with $\delta \le (1 - \alpha)^2$,

$$\max\{D_h(x, \partial B_{\delta r}(x)), D_h(\text{around } A_{\delta r,\sqrt{\delta} r}(x))\} \le \delta^\theta D_h(u, v). \tag{17}$$

Only countably many scales are needed for endpoint recovery.

Proposition 4.2. *For every $K \Subset U$, $\alpha \in (3/4, 1)$, $\theta > 0$, and $\eta > 0$,*

$$\sup_{z \in K} \mathbb{P}\big[\textit{some regular pair at } (z, r) \textit{ has } e(u, v) > (1 + \eta) D_h(u, v)\big] \longrightarrow 0 \quad (r \to 0). \tag{18}$$

Proof. Otherwise choose $r_n \to 0$ and $z_n \to z \in K$ with failure probabilities bounded below. Rescale the field and map by

$$\widehat{h}_n = h(z_n + r_n\,\cdot) - (h(z_n + r_n\,\cdot), \rho), \qquad \psi_n(w) = \frac{\phi(z_n + r_n w) - \phi(z_n)}{r_n}.$$

The fields $\widehat{h}_n$ have the same smoothly normalized GFF law and $\psi_n'(0) \to \phi'(z) \neq 0$. Both metric coordinates are divided by $r_n^{\xi Q} \exp\{\xi(h(z_n + r_n\,\cdot), \rho)\}$. Confine them to expanding closed disks. Proposition 2.3 still compares the confined distances. The source geodesic of a regular pair lies in the unit annulus, so its distance is unchanged by confinement; a target comparison failure persists because confinement can only increase the target distance.

The normalized upper bound in (16) is a tight log-normal multiple of $(1-\alpha)^{-1}$. Restrict first to normalized source distances at most M. All endpoint positions then lie in a fixed compact set and their separation is at least $1 - \alpha$. Failure witnesses can be selected measurably after completion, as explained in Appendix A.2.

Adjoin the selected endpoints, the source distance, and the indicator of failure to the joint convergence of Proposition 3.4. The last three coordinates are tight. Pass to a subsequence and use a Skorokhod representation in the ambient separable state spaces. The limiting metric coordinates are equal almost surely. On any represented sample where failure occurs infinitely often and the source distances are at most M, pass to a further samplewise subsequence with convergent endpoints and source distances.

For every fixed small dyadic δ, the bounds (17) remain valid for the confined metrics once their carriers contain the corresponding balls. For point-to-circle distance, stop a near-minimizing path at its first exit; for around-distance, the loop is already in the small annulus. Lemma 4.1 therefore applies and gives

$$(1 + \eta)L \leq \limsup_n e_n(u_n, v_n) \leq D(u, v) \leq L.$$

Here $L \geq D(u, v) > 0$, since $u \neq v$. This is a contradiction. Hence the failure probability with the cutoff M tends to zero. Let $M \to \infty$. Since the argument treats every possible sequence of centers in K, it proves the uniform statement. □

5 Good annuli along geodesics

The comparison in Proposition 4.2 is useful only if regular segments occupy a definite fraction of a long geodesic. We now construct the annuli which ensure this. All geometric estimates in the first two subsections concern the source metric alone.

5.1 Small loops near a geodesic

Put

$$\mathcal{A} = \overline{B}_{2r}(z) \setminus B_{r/2}(z), \qquad \mathcal{A}_0 = \overline{B}_{3r/2}(z) \setminus B_{3r/4}(z).$$

A boundary-to-boundary geodesic in $\mathcal{A}$ is a geodesic for the metric internal to this closed carrier, with both endpoints on $\partial\mathcal{A}$. The endpoints may lie on the same boundary component.

Lemma 5.1. *There is $\theta > 0$ such that, with probability tending to one as $q \to 0$, the following holds simultaneously for every boundary-to-boundary geodesic P in $\mathcal{A}$, every $\delta \in \Delta$ with $\delta \leq q$, and every $x \in \mathcal{A}_0$ for which $P \cap B_{\delta r}(x) \neq \varnothing$:*

$$D_h(\text{around } A_{\delta r, \sqrt{\delta} r}(x)) \leq \delta^{\theta} D_h(\text{across } A_{\delta r, \sqrt{\delta} r}(x)). \tag{19}$$

The probability bound is uniform in z, r.

Proof. We prove a bound $O(\delta^b)$ for failure at one scale, with $b > 0$, and then sum over dyadic scales. Rescale to $z = 0, r = 1$; all ratios are invariant under the resulting random metric normalization.

Take a lattice of mesh $\delta/100$ covering a fixed neighborhood of $\mathcal{A}_0$. Around each lattice point w, use annuli $A_{t,4t}(w)$ with radii separated by a factor of ten, between 10δ and $\sqrt{\delta}/10$. There are $N \geq c\log(1/\delta) - O(1)$ such annuli. By annular tightness, for a sufficiently large deterministic M each has probability arbitrarily close to one of satisfying

$$D_h(\text{around } A_{2t,3t}(w)) \leq M D_h(\text{across } A_{t,4t}(w)). \tag{20}$$

Annular iteration, followed by the union bound over $O(\delta^{-2})$ lattice points, shows that with probability $1 - O(\delta^b)$ at least $N/2$ of these annuli are good at every lattice point. The iteration exponent can be chosen large enough to pay for this lattice count.

Fix a geodesic and a point x as in the lemma, and choose w within $\delta/50$ of x. All the test annuli lie between the two circles centered at x, and their surrounding loops separate $B_\delta(x)$ from the exterior of $B_{\sqrt{\delta}}(x)$. They also lie strictly inside $\mathcal{A}$. List the good annuli in increasing order of radius. Write a_j for their crossing costs and choose loops of lengths $\ell_j \leq 2Ma_j$. Every a_j is positive and finite.

The geodesic enters the innermost ball, while both its endpoints lie outside all these annuli. Between its first and last hits of loop j, it crosses each smaller test annulus twice. By geodesic minimality, replacing this portion by an arc of loop j gives

$$2\sum_{i<j} a_i \leq \ell_j \leq 2Ma_j.$$

Consequently, if $S_j = \sum_{i\leq j} a_i$, then $S_j \geq (1 + M^{-1})S_{j-1}$. Any path crossing the full annulus centered at x must cross every test annulus, so its cost is at least S_m, where $m \geq N/2$. The first good loop is an admissible loop around that full annulus. Hence

$$\frac{D_h(\text{around } A_{\delta,\sqrt{\delta}}(x))}{D_h(\text{across } A_{\delta,\sqrt{\delta}}(x))} \leq 2M(1 + M^{-1})^{-(m-1)}.$$

Choose $\theta > 0$ smaller than the power of δ on the right. For all sufficiently small δ, this is at most δ^θ. Finally, $\sum_{\delta\in\Delta,\,\delta\leq q} O(\delta^b) = O(q^b)$. The construction is simultaneous in the geodesic and its nearby center, which proves the stated uniformity. □

The ratio on the right of (19) is essential: the denominator is the crossing distance of the same annulus. The proof above obtains it by summing the costs of disjoint crossings and using geodesic minimality at each surrounding loop.

5.2 Geodesics do not follow a thin circular annulus

Lemma 5.2. *Fix $q > 0$. As $\alpha \uparrow 1$, with probability tending to one, every segment of every boundary-to-boundary geodesic in $\mathcal{A}$ which is contained in $\overline{A}_{\alpha r,r}(z)$ has Euclidean diameter at most qr. The convergence is uniform in z, r.*

Proof. Again take $z = 0, r = 1$. We describe the local obstruction which prevents a geodesic from following the unit circle. At a fixed point of that circle and a fixed scale t, there is a probability $p_0 > 0$, independent of the scale, of finding a loop in $A_{2t,3t}$ with length less than the crossing distance of $A_{t,3t/2}$. To see the positive probability, first restrict to an event where the normalized crossing cost is bounded below and the normalized loop cost is bounded above. It has positive

probability by annular tightness. Subtract a sufficiently large smooth bump which is constant on the outer loop annulus and zero on the inner crossing annulus. Weyl scaling makes the loop cheaper than the crossing. Cameron–Martin absolute continuity then gives positive probability for the unshifted field. The Dirichlet energy of the rescaled bump is constant, so the lower bound is uniform in scale.

Apply annular iteration to scales between ε and $\sqrt{\varepsilon}/10$. With probability $1-O(\varepsilon^b)$, at least one such shortcut loop surrounds $B_\varepsilon(x)$, for each fixed point x on the circle. A boundary-to-boundary geodesic of $\mathcal{A}$ cannot enter this ball: its endpoints are outside the loop, and the segment between its first and last intersections would cost more than an arc of the loop. All loops used lie in the interior of $\mathcal{A}$, so this conclusion also holds for geodesics whose endpoints lie on its boundary.

Parametrize the unit circle by arclength. Fubini's theorem and Markov's inequality show that, outside an event of probability $O(\varepsilon^{b/2})$, the total arclength of points whose ε-balls can be visited by any such geodesic is at most $O(\varepsilon^{b/2})$. This statement is simultaneous in the geodesic, because the shortcut obstruction is deterministic once its loop exists.

If a connected path lies in $\overline{A}_{1-\varepsilon/2,1}$ and has diameter greater than q, its radial projection onto the unit circle covers an arc of length at least cq, for small ε depending on q. Every point of this arc is within ε of the path. This contradicts the preceding bound when ε is small. Taking $\alpha \geq 1-\varepsilon/2$ proves the lemma. □

5.3 The good event

Fix the comparison constant C from Proposition 2.3 and set $k = (4C^2)^{-1}$. Let

$$S = D_h(\mathcal{A}_0, \partial\mathcal{A}; \mathcal{A}).$$

This barrier cost is positive. Given $p < 1$, we may choose a dyadic $q \in (0, 1/100)$, then $\alpha \in (1-q, 1)$, and then $A_0 > 1$ so that the following four conditions hold jointly with probability at least p, uniformly in z, r:

(G1) The conclusion of Lemma 5.1 holds for all dyadic $\delta \leq q$ and all the specified internal geodesics.

(G2) The outer loop cost satisfies

$$D_h(\text{around } A_{3r/2,2r}(z)) \leq \min\{(1-\alpha)^{-1} m_{z,r},\ kq^{-\theta} S\}. \tag{21}$$

(G3) Every segment of a boundary-to-boundary geodesic in $\mathcal{A}$ contained in $\overline{A}_{\alpha r,r}(z)$ has diameter at most qr.

(G4) The thin annulus satisfies

$$D_h(\text{around } A_{\alpha r,r}(z)) \leq A_0 D_h(\text{across } A_{\alpha r,r}(z)). \tag{22}$$

Here is the order of the choices. Lemma 5.1 makes the failure probability of (G1) small when q is small. The ratio of the outer loop cost to S has a tight law, independent of z, r, so decreasing q also gives the second bound in (G2). Next choose α sufficiently close to one for (G3), using Lemma 5.2, and for the first bound in (G2), using annular tightness. Finally, for this fixed α, increase A_0 to obtain (G4). Assigning a quarter of the allowed failure probability to each condition proves the joint assertion.

For $\eta > 0$, add one more condition:

(G5) Whenever a regular pair at (z, r) satisfies

$$|u - v| \le qr, \qquad D_h(\text{around } A_{qr,\sqrt{q}r}(v)) \le kS, \tag{23}$$

it also satisfies $e(u, v) \le (1 + \eta)D_h(u, v)$.

Denote this five-condition event by $F_{z,r}$.

Lemma 5.3. *For every $p < 1$ the parameters q, α, A_0 can be chosen independently of η so that, for every $K \Subset U$ and $\eta > 0$,*

$$\inf_{z \in K} \mathbb{P}[F_{z,r}] \ge p$$

for all sufficiently small r. The event is determined by the field on a fixed annular collar of $\mathcal{A}$, modulo an additive constant.

Proof. Choose (G1)–(G4) with probability strictly larger than p. Proposition 4.2 shows that the additional failure probability of (G5) tends uniformly to zero. These choices do not involve η until this last scale cutoff.

For locality, only (G5) needs an explanation. Under (23), there is a loop surrounding both u, v of source length less than $2kS$. It lies in the middle of $\mathcal{A}$. Its target length is less than $2CkS = S/(2C)$, whereas a target path from this middle region to $\partial\mathcal{A}$ costs at least S/C. Any excursion out of $\mathcal{A}$ can therefore be replaced by an arc of the loop, for both metrics. Thus both distances between u, v are computed internally in $\mathcal{A}$. The same observation identifies a geodesic specified in (G5) with an internal one.

All the endpoint circles and loops in the regularity tests lie inside $\mathcal{A}$. A point-to-circle test uses a path stopped at its first exit. The five conditions can consequently be read from the two internal metrics and the circle average on this annular collar. By domain locality the target internal metric is a function of h^ϕ on the image collar, hence of h on the source collar. Adding a constant multiplies every cost and $m_{z,r}$ by the same factor. The event is therefore invariant under constants. Appendix A.2 justifies completion of the sigma algebra when endpoint quantifiers occur. □

Lemma 5.4 (Good covers)**.** *Fix $K \Subset U$ and $\eta > 0$. Almost surely there is a deterministic sequence $\varepsilon_j \downarrow 0$ along which every $x \in K$ belongs to some $B_{r/2}(z)$ with $r \in [\varepsilon_j^2, \varepsilon_j]$ and $F_{z,r}$ occurring. One sequence can be used for a countable compact exhaustion.*

Proof. Use lattice centers of mesh $\varepsilon^2/100$ in a compact neighborhood of K, and geometrically separated radii between ε^2 and ε. Choose the probability in Lemma 5.3 sufficiently close to one that annular iteration gives failure probability $O(\varepsilon^{10})$ at each center. The union bound over $O_K(\varepsilon^{-4})$ centers gives failure probability $O_K(\varepsilon^6)$. A center within $\varepsilon^2/50$ of x has $x \in B_{r/2}(z)$ at every tested radius. Choose a summable sequence of errors and use Borel–Cantelli, with a diagonal choice over the exhaustion. □

6 Proof of Theorem 1.3

6.1 Improving the comparison constant

We first prove $e \le d$ for a whole-plane GFF restricted to U. Let

$$C_* = \inf\{c \ge 1 : \ \mathbb{P}[e(z,w) \le c\, d(z,w) \text{ for all } z, w \in U] = 1\}.$$

Proposition 2.3 gives $C_* \le C < \infty$. Intersecting the events for a decreasing sequence of admissible constants shows that the bound also holds at C_*. Suppose $C_* > 1$ and choose $0 < \eta < C_* - 1$. Keep the parameters of the good events fixed and put

$$C_{\text{new}} = (1+\eta) + \frac{A_0}{A_0+1}(C_* - 1 - \eta). \tag{24}$$

Then $C_{\text{new}} < C_*$. We will prove that this smaller constant also compares the metrics.

Lemma 6.1 (An improved geodesic step)**.** *Let $P : [0,T] \to U$ be a whole-plane D_h-geodesic parametrized by length. Suppose $F_{z,r}$ occurs, both global endpoints of P are outside $\overline{B}_{2r}(z)$, and $P(a) \in B_{r/2}(z)$. Let $t > a$ be the next exit time from $B_r(z)$, and let s be the last time in $[a,t]$ at which $P(s) \in \partial B_{\alpha r}(z)$. Then*

$$e(P(s), P(t)) \le (1+\eta)(t-s), \tag{25}$$

$$s - a \le A_0(t-s). \tag{26}$$

Consequently,

$$e(P(a), P(t)) \le C_{\text{new}}(t-a). \tag{27}$$

Proof. The segment $P|_{[s,t]}$ lies in the closed thin annulus. Extend it backwards to the last preceding boundary hit of $\mathcal{A}$, and forwards to the next boundary hit. This gives a boundary-to-boundary internal geodesic to which (G1)–(G3) apply. The global endpoints being outside $B_{2r}(z)$ also ensure that P crosses any outer loop before and after this segment. Geodesic minimality and (G2) yield

$$t - s \le (1-\alpha)^{-1} m_{z,r}.$$

Condition (G3) bounds the diameter of $P|_{[s,t]}$ by qr.

Write $u = P(s)$ and $v = P(t)$. For a dyadic $\delta \le (1-\alpha)^2$, the segment from u to v crosses $A_{\delta r, \sqrt{\delta} r}(u)$, since $|u - v| \ge (1-\alpha)r$. Its crossing cost is at most $t - s$. Condition (G1) therefore bounds the surrounding-loop cost by $\delta^\theta(t-s)$. The full geodesic crosses this loop on both sides of u; replacing the intervening piece by the loop shows that its portion from u to $\partial B_{\delta r}(u)$ has length at most the same loop cost. Taking infima proves both bounds in (17). The argument at v is identical. Thus (u, v) is a regular pair.

Apply (G1) at scale q around v. The relevant crossing is part of the portion of P enclosed by an outer loop. Minimality and (G2) give

$$D_h(\text{around } A_{qr, \sqrt{q} r}(v)) \le q^\theta D_h(\text{around } A_{3r/2, 2r}(z)) \le kS.$$

The diameter bound supplies the other condition in (23). Condition (G5) now proves (25).

To prove (26), take a near-minimizing loop around the thin annulus. It meets P before time a and between s and t. The part of P between those hits has length at least $s - a$ and, by minimality, at most the loop length. Hence, by (G4),

$$s - a \le D_h(\text{around } A_{\alpha r, r}(z)) \le A_0 D_h(\text{across } A_{\alpha r, r}(z)) \le A_0(t-s).$$

All statements with an infimum follow by letting the loop's length error tend to zero.

The old comparison gives $e(P(a), P(s)) \le C_*(s-a)$. Combine it with (25), and use $(s-a)/(t-a) \le A_0/(A_0+1)$. This gives exactly (27). □

Now take any finite whole-plane geodesic $P : [0, T] \to U$ with compact image. If $T = 0$ there is nothing to prove. Otherwise P is injective, Euclidean continuous, and a homeomorphism onto its image. Work on the probability-one event of good covers for a compact neighborhood of that image.

At a cover scale ε, retain the corridor between the last exit from $B_{4\varepsilon}(P(0))$ and the first subsequent entry into $B_{4\varepsilon}(P(T))$. Its endpoint times tend to 0 and T. Every retained point is at distance at least 4ε from both global endpoints. Therefore every good ball $B_r(z)$ with $r \le \varepsilon$ containing that point in its inner half has both global endpoints outside $B_{2r}(z)$, as required in Lemma 6.1.

Starting at the beginning of this corridor, choose a good ball and take its next exit time from $B_r(z)$. Repeat from that time until reaching the end of the corridor. At each complete step, Lemma 6.1 bounds the target cost by C_{new} times the source time. There are finitely many complete steps: the finite covering collection has a positive minimum crossing cost. If the final chosen ball contains the rest of the corridor, leave that last partial step unimproved. Its image has diameter at most 2ε, so its time length tends to zero by uniform continuity of the inverse of P.

The initial and terminal discarded portions also have time length tending to zero. The old constant C_* bounds their target costs. Summing the complete steps and sending $\varepsilon \to 0$ gives

$$e(P(0), P(T)) \le C_{\text{new}} T. \tag{28}$$

The good-cover event was independent of P, so this conclusion holds simultaneously for every such geodesic.

Finally let Q be any finite-length source path in U. Choose a compact neighborhood $K \Subset U$ of its image. The source distance from the image to ∂K is positive. Partition Q into pieces shorter than this distance, and replace each piece by a whole-plane geodesic between its endpoints. Each replacement stays in K, since leaving K would already cost more than the piece it replaces. Apply (28) to these replacements, sum, and infimize over Q. This proves $e \le C_{\text{new}} d$ at every endpoint pair, contradicting the definition of C_*. We conclude that

$$e \le d. \tag{29}$$

6.2 Other fields and the inverse map

We transfer (29) to the domain fields of the theorem. Cover U by countably many disks $B \Subset B' \Subset U$. On B' couple the zero-boundary part of h to that of a whole-plane GFF H. On B the difference $h - H$ is a random continuous function g. The corresponding target difference is $g \circ \phi^{-1}$.

The one-sided comparison for H compares lengths of every path. Applying the common Weyl weight preserves this inequality: for a path parametrized by source length, the target length measure is bounded by the source length measure, so the inequality persists after integration against the positive continuous function $e^{\xi g}$. Subdividing into small pieces proves the same statement without choosing a particular parametrization. Thus (29) holds for the internal metrics on B associated with h. The simultaneous Weyl event permits dependence between g and H.

Intersect the events for the countable disk cover. Any finite-length path in U has a finite subdivision subordinate to this cover. Summing the local inequalities and infimizing gives (29) for the full domain metric of h.

Apply this conclusion on V to h^ϕ and ϕ^{-1}. The transformed field is again a zero-boundary GFF plus a continuous function. The chain rule gives

$$(h^\phi)^{\phi^{-1}} = h + Q \log |(\phi^{-1})' \circ \phi| + Q \log |\phi'| = h.$$

The inverse inequality is $d \le e$. Together with (29), it proves Theorem 1.3 on any extension used to realize the fields. The metric functions depend only on the original fields, so the identity descends to their original probability space, as shown in Appendix A.1. All inequalities are between nonnegative extended distances; the argument never subtracts infinite values and therefore includes singular endpoints.

Proof of Corollary 1.4. Choose product realizations for the countable list of source fields and for their transformed fields. Apply the positive-weight mixture construction in Lemma 2.1 separately to the two lists. This gives one pair of measurable functions Φ_U, Φ_V. For each index, Theorem 1.3 applies with these versions and gives the equality on its original probability space. Realization independence on common extensions proves the final assertion. If the fields share one probability space, their countably many full-probability events can also be intersected. □

A Measurable versions and endpoint selection

A.1 Versions on the original probability space

This appendix supplies the measure-theoretic details used in the main text. They do not alter the geometric argument.

Equip $\mathcal{D}'(U)$ with its standard measurable structure, generated by test-function evaluations. Compact internal metrics are encoded by their closed epigraphs. On an exhaustion $K_n \subset \operatorname{int} K_{n+1}$, a countable sequence of these encodings determines the domain metric. This gives a standard Borel coordinate space for the construction, even when the open-domain internal distance is not itself lower semicontinuous at the boundary of a confinement set.

For a fixed law of h, construct a product realization over its original space $(\Omega, \mathcal{F}, \mathbb{P})$ as in Lemma 2.1. Locality on a countable cover and the common-function argument identify every compact metric coordinate as a measurable function of h. Measurable factorization therefore yields a function Φ_U with

$$\Phi_U(h) = D^U_{H|_U + F}$$

almost surely on that product realization, simultaneously in all endpoints. Extend the function arbitrarily off the full-measure set on which the required metric coordinates are consistent. Its coordinate evaluations are measurable.

This version works on every measure-preserving extension $\pi : \Omega' \to \Omega$ carrying a realization of $h \circ \pi$. Indeed, sample the auxiliary whole-plane coordinate of the fixed product realization by its conditional kernel given $\omega = \pi(\omega')$. This produces a common refinement of the two extensions. The conditional kernel exists because the auxiliary distribution coordinate is standard Borel; no regular conditional probability on Ω itself is required. On the refinement, pull back the defining identity for Φ_U and use realization independence to identify the two metrics. The identity thus holds almost surely there.

If E is the event that the required identity holds on Ω', its preimage under the measure-preserving projection has probability one. Therefore E itself has probability one. The same argument descends the final covariance identity to Ω. This proves both the original-space and all-realizations clauses of the main results, and also shows that existence of realizations is part of the construction, rather than an implicit assumption.

For a countable family of laws, give each member positive weight and carry out the preceding construction under the mixture. A set of mixture measure zero has zero measure under every component. The resulting functions therefore work for each law separately. This is the only uniformity in field laws needed in the paper.

A.2 Selecting a failing pair

The probabilities in Proposition 4.2 include an existential quantifier over uncountably many endpoints. Here is a precise way to interpret that quantifier.

Use the underlying fields, their countably many compact metric coordinates, endpoints in a compact set, and a path in $C([0,1],K)$ as coordinates in standard Borel spaces. For a lower-semicontinuous metric and a Euclidean-continuous path, its length is the supremum of the sums of distances over partitions with rational times. Indeed, any finite partition can be approximated by rational ones, and lower semicontinuity gives the required inequality. Thus length is a measurable function of these coordinates. A path is a geodesic precisely when it joins the specified endpoints and its length equals their distance.

Point-to-circle and around distances can also be included as measurable coordinates. One way to verify this is to use paths on compact carriers with prescribed rational bounds for their metric speed and their Euclidean modulus. These path families are compact. The length sublevel relation is closed under uniform path limits and epigraph lower bounds for the metrics. For a surrounding loop, use a rational positive collar away from the inner and outer boundaries; its separation property is preserved under uniform limits within that collar. Taking countable unions over the collars, length bounds, and moduli recovers the relevant infima. Moving centers and radii are added as ordinary Euclidean coordinates. The local metric-to-Euclidean modulus ensures that every finite-length test path is covered by one of these countable families.

Consequently the relation asserting a source geodesic, (16), the countably many tests (17), and the strict comparison failure is Borel after the path is included. Its projection to the field and metric coordinates is analytic, hence universally measurable. The measurable selection theorem for analytic relations supplies witnessing endpoints after completion of the probability space. Only these endpoints, the bounded source distance, and a binary failure mark are adjoined in the subsequence argument; a path coordinate need not be made tight.

These observations also apply to the local annular events. For any one field law, choose measurable versions of their indicators in the completed local sigma algebra. Only countably many centers and scales enter a cover, so all versions can be fixed on one full-probability event before annular iteration and Borel–Cantelli are applied.

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